MIL-E-44141(GL)
28 March 1985

MILITARY SPECIFICATION

ESCALLOPED POTATOES WITH PORK, COOKED, DEHYDRATED -

(OPERATIONAL RATION COMPONENT)

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers cooked, dehydrated escalloped potatoes with pork for use by the Department of Defense as a component of the Food Packet, Assault.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-44146 - Packaging of Food Packet Assault Components in Flexible Bags

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Standards for Grades of Frozen Sweet Peppers
U.S. Standards for Grades of Canned Pimientos
U.S. Standards for Sweet Peppers

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Instant Nonfat Dry Milk
U.S. Standards for Grades of Cheddar Cheese

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(Application for copies should be addressed to the Director, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

THE UNITED STATES PHARMACOPOEIAL CONVENTION (USP)

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney Street, No. 410, San Francisco, CA 94108.)

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NATIONAL CHEESE INSTITUTE

Cheese Color Reference Standards

(Application for copies should be addressed to the National Cheese Institute, 110 N. Franklin Street, Chicago, IL 60606.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Pork. Pork shall be the only meat ingredient used and shall be derived from any one or any combination of the following recognizable bone-in wholesale cuts: shoulders, shoulder picnics, Boston butts, hams and loins. Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All pork destined for use in the product shall be in excellent condition, i.e. exposed lean and fat surfaces shall be of a color and bloom normally associated with the class and cut of meat and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum packaged meat in excellent condition shall be acceptable. Also, the pork shall show no evidence of freezing, defrosting or mishandling. Coarse textured dark pork as typically produced from aged sows, stags, or boars and pale, soft and exudative (PSE)

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pork is not acceptable. Cuts which have had a slight amount of bruised lean fully removed will be acceptable. Cuts which have had more than a slight amount of lean removed for any reason shall not be included. All suitable lean meat shall be used in proportions as existing in the involved cut except that the tenderloins may be excluded at the option of the contractor.

3.2.1.1 Boning and trimming. The pork shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed pork shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed pork prior to processing into the cooked pork logs shall be in accordance with the following requirements:

a. Pork processed on the same day as boning and trimming shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for not more than 4 days is permitted provided that the pork is maintained in the temperature range of 28° to 40°F (inclusive).

3.2.2 Potatoes. Potatoes shall be fresh and shall be a variety of excellent processing qualities and specific gravity of 0.95 to 1.10. Varieties with these characteristics include Russet, Burbank, Kennebec, Monona, and Norchips. The potatoes shall be washed, sorted, peeled, trimmed to remove defects and mechanically diced before cooking with dicer settings at approximately 3/8 by 3/8 by 3/8 inch. Dehydrated or dehydrofrozen potatoes shall not be used.

3.2.3 Peppers, green and red, sweet. Sweet green peppers shall be fresh or frozen and shall meet the requirements of either 3.2.3.1 or 3.2.3.2. Sweet red peppers shall be fresh, frozen, or canned and comply with the requirements of 3.2.3.3, 3.2.3.4 or 3.2.3.5. The peppers shall be trimmed, cored and mechanically diced with dicer settings at approximately 1/4 by 1/4 by 1/4 inch.

3.2.3.1 Peppers, green, sweet, fresh. Green sweet peppers shall be fresh, clean, sound, firm pods of a single variety or of similar varietal characteristics. The peppers shall be U.S. No. 1.

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3.2.3.2 Peppers, green, sweet, frozen. Frozen green sweet peppers shall be U.S. Grade B or better of the U.S. Standard for Grades of Frozen Sweet Peppers.

3.2.3.3 Peppers, red, sweet, fresh. Red sweet peppers shall be fresh, clean, sound, firm and free from rot or mold. The peppers shall be well colored (not less than 90 percent of the surface area shall have a characteristic medium or dark red color and the remainder of the surface shall not be dominated by green).

3.2.3.4 Red peppers, frozen. Frozen red peppers shall be U.S.Grade B or better of the U.S. Standards for Grades of Frozen Sweet Peppers.

3.2.3.5 Red peppers, canned. Canned red peppers shall be U.S. No. 1 in accordance with the U.S. Standards for Sweet Peppers prior to canning.

3.2.4 Pimientos. Pimientos shall be diced style pieces and shall meet the requirement for grade C or better of the U.S. Standards for Grades of Canned Pimientos.

3.2.5 Milk, nonfat, dry, instant. Instant nonfat dry milk shall meet the requirements for U.S. Extra Grade, Low Heat, as defined in the U.S. Standards for Instant Nonfat Dry Milk.

3.2.6 Starch, freeze-thaw-stable. Freeze-thaw-stable starch shall form a smooth, short textured, heavy bodied, almost clear aqueous solution. The flavor of the cooked solution shall be bland with essentially no cereal or starchy taste.

3.2.7 Flour, wheat. The wheat flour shall be free flowing, enriched general purpose white (bleached), free from mold, and free from lumps that do not break under light finger pressure.

3.2.8 Cheese, cheddar, aged, dehydrated, yellow. Dehydrated cheddar cheese shall be derived from U.S. Grade A or better aged cheddar cheese of the U.S. Standards for Grades of Cheddar Cheese. The dehydrated cheese shall be uniform in composition and appearance, free from lumps that do not fall apart under light finger pressure, and free from burnt specks and foreign material. The cheese shall be free from rancid flavors and odors. The color shall be within the range of No. 5 to No. 8 level when compared to the National Cheese Institute's Cheese Color Reference Standards. The

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dehydrated cheese shall contain not more than 3.0 percent moisture, not more than 5.0 percent salt and not less than 48.0 percent milkfat.

3.2.9 Salt. Salt shall be non-iodized, white, refined, sodium chloride, with or without anti-caking agents.

3.2.10 Onion, dehydrated, chopped. The dehydrated chopped onion shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.11 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L., from which the outer covering or outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot-biting-pungent flavor, and a light color. Volatile oil content shall be not less than 1.0 mL/100g and the white pepper shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.12 Nutmeg, ground. The ground nutmeg shall be derived from the dried seeds of Myristica fragrans Houtt and shall have a dull brown color. Volatile oil content shall be not less than 7.5 mL/100g and the nutmeg shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.13 Water. Water used for formulation, ice making and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.14 Nitrogen. Nitrogen shall be U.S. Pharmacopoeia grade and shall be water or liquid nitrogen pumped.

3.3 Preparation and processing.

3.3.1 Preparation of the pork dices. The pork dices shall be prepared in accordance with 3.3.1.1, 3.3.1.3 and 3.3.1.4 or 3.3.1.2, 3.3.1.3 and 3.3.1.4. Mixing and stuffing shall be continuous and shall be completed within one day.

3.3.1.1 Grinding. Boneless pork shall be ground once through a plate having openings 1 to 1-1/2 inches in diameter. The ground pork shall be mixed with 0.5 percent salt until moderately tacky. The pork shall be mechanically tightly stuffed into suitable casings for cooking. The pork shall not exceed 50°F and shall have a fat content of 9 to 16 percent.

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3.3.1.2 Flake-cutting. Boneless pork shall be flake-cut by passing it once through an Urschel Comitrol equipped with a 3J030750D head or equivalent. The flake-cut pork shall be mixed with 0.5 percent salt until moderately tacky. The pork shall be mechanically tightly stuffed into suitable casings for cooking. The pork shall not exceed 40°F after flake-cutting (dry ice is permitted to maintain temperature) and shall have a fat content of 9 to 16 percent.

3.3.1.3 Cooking. The logs (see 3.3.1.5 b for chilling and freezing conditions) shall be cooked in water or steam until an internal temperature of not less than 155°F and not greater than 165°F is reached.

3.3.1.4 Dicing. The cooked pork logs at a uniform internal temperature of 38°F or lower (see 3.3.1.5 c, d and e for chilling and freezing conditions) shall be mechanically diced with dicer settings at approximately 1/2 by 1/2 by 1/4 inch. After dicing, there shall be a minimum of 75 percent recognizable dices (approximately 1/2 by 1/2 by 1/4 inch).

3.3.1.5 Time-temperature limitations. The pork shall be handled so as to comply with the following limitations:

a. Maximum holding temperature and time of uncooked, chilled boneless pork:

	50°F	24 hours

b. Maximum holding temperature and time of uncooked pork logs:

1. If pork logs are chilled	40°F	24 hours
2. If pork logs are frozen 1/	0°F 2/	30 days

c. Maximum holding temperature and time of cooked chilled pork logs for dicing: 38°F 3/ 48 hours

d. Maximum holding temperature and time of cooked, frozen pork logs for dicing: 0°F 4/ 21 days

e. Maximum holding temperature and time of cooked, tempered (see 6.3) pork logs for dicing: 38°F 48 hours

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f. Maximum holding temperature and time of cooked pork dices:

1. If cooked pork dices are chilled	40°F	4 hours
2. If cooked pork dices are frozen 5/	0°F 4/	14 days

1/ If pork logs are frozen in the uncooked state they shall not be frozen after cooking and shall be used in the product within 24 hours after chilling.

2/ Uncooked pork logs shall be frozen to 0°F within 72 hours after placing in the freezer.

3/ Cooked pork logs shall be chilled to 38°F within 8 hours after cooking.

4/ Cooked pork logs and cooked pork dices shall be placed in a freezer within 4 hours after completion of cooking and frozen to an internal temperature of 0°F or below within 48 hours after placing in the freezer.

5/ Pork dices prepared from pork logs that have been frozen prior to cooking or cooked pork logs that have been frozen prior to dicing shall not be frozen.

3.3.2 Preparation of the escalloped potatoes with pork. The escalloped potatoes with pork shall be prepared as preconcentrated and formed slices or as compressed bars. Slice preparation shall be as outlined by 3.3.2.1, 3.3.2.2, and 3.3.2.3. Bar preparation shall be as outlined by 3.3.2.1, 3.3.2.2 and 3.3.2.4.

3.3.2.1 Formulation of the escalloped potatoes with pork. The escalloped potatoes with pork shall be formulated as follows:

Ingredient	Percent by weight
Potatoes, raw, diced	33.5
Water 1/	29.0
Pork, cooked, diced	26.0
Cheese sauce mix (see below)	3.85
Milk, nonfat, dry, instant	3.5
Peppers, green, diced	1.7
Peppers, red, diced or pimientos	1.7
Starch	0.5
Flour	0.25

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Cheese sauce mix

Ingredient	Percent by weight
Cheddar cheese, dehydrated	62.3
Salt	18.2
Onions, chopped, dehydrated	16.4
White pepper, ground	1.8
Nutmeg, ground	1.3

1/ For preconcentrated and formed slices the formula water shall be reduced by an amount that will yield a product complying with the finished product requirements.

3.3.2.2 Mixing and cooking. The escalloped potatoes with pork shall be mixed and cooked as follows:

a. The raw diced potatoes shall be cooked in boiling water until the potatoes have firm edges and a slightly raw flavor. The potato dices shall be rinsed with cold water to stop cooking and remove starch.

b. The instant, nonfat dry milk shall be reconstituted using part of the water.

c. A slurry shall be prepared with the starch, flour and part of the water.

d. The cheese sauce mix ingredients and the remaining water shall be added to the slurry and mixed while heating to 160°F. The sauce shall be mixed until smooth.

e. The green peppers and red peppers (or pimientos) and diced, cooked pork shall be added and heated to 160°F.

f. The cooked potatoes and reconstituted nonfat dry milk shall be added and mixed gently.

3.3.2.3 Slice preparation. The preconcentrated and formed slices shall be prepared as follows; or alternatively, the cooked product may be formed so that the dimensions, weights and volumes specified in 3.4(e) are met. The times and temperatures specified below shall be followed according to the process being used.

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a. The cooked product shall be immediately mechanically stuffed into suitable casings. Casings shall have an approximate 5-1/2 inch lay-flat dimension.

b. Each stuffed casing shall be formed (see 6.4) into a rectangular shape in accordance with 3.4(e).

c. The product shall be placed in a freezer within 4 hours after stuffing and frozen to an internal temperature of 0°F or below within 24 hours after placing in the freezer. Frozen product shall not be held for longer than 30 days before slicing and freeze dehydration.

d. The frozen product shall be cut into slices so that four dehydrated slices will meet the requirements specified in 3.4 (e).

e. The slices shall be placed in the freeze dryer and dehydrated as specified in 3.3.3. The dehydration shall be started within 24 hours after slicing. The product shall be protected at all times from contamination and freezer burn.

3.3.2.4 Bar preparation. The compressed bars shall be prepared as follows:

a. The cooked product shall be portioned onto freeze dryer trays.

b. The product shall be placed in a freezer within 4 hours after cooking and frozen to 0°F or lower within 12 hours after placing in the freezer. The product shall be protected at all times from contamination and freezer burn.

c. The product shall be placed in the freeze dryer and dehydrated as specified in 3.3.3. The dehydration shall be started within 48 hours after placing in the freezer.

d. The dehydrated product shall be conditioned by spraying with 10 percent by weight of water and held under vacuum or nitrogen for not less than 8 hours. Spray an additional 3 percent by weight of water immediately before compression. Microwave conditioning shall not be used for this product.

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e. The conditioned product shall be compressed (at 400 to 450 psi) into bars 1 inch wide by 3 inches long by not more than 1-1/4 inches thick so that the dehydrated bars will meet the requirements specified in 3.4 (f). The compressed bars shall have edges and corners rounded to a nominal 1/8 inch curvature. The product shall be compressed within 5 days of initial freeze dehydration.

f. The compressed bars shall be placed in the freeze dryer and dehydrated as specified in 3.3.3. The dehydration shall be started within 48 hours of compression.

3.3.3 Dehydration. The escalloped potatoes with pork shall be freeze dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury and at a product temperature as indicated by suitable instruments, not to exceed 130°F. If the platen temperature is maintained at 135°F or below with radiant heating, the product temperature may be disregarded. After dehydration is completed, the pressure shall be equalized to atmospheric level with nitrogen. In no case shall more than 16 hours elapse between dehydration and completion of packaging. During this interim period product shall be adequately protected from oxygen and moisture by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or under a vacuum of at least 27 inches of mercury for the entire period. If vacuum is used, it shall be broken with nitrogen. For compressed bar product, during the interim between freeze drying and compression the product shall also be protected as above.

3.3.4 Pouch filling and sealing. Each pouch shall be filled and sealed such as to conform to the following requirements:

a. Three bars or four slices shall be inserted into a mixing bag conforming to the requirements of MIL-P-44146.

b. The bag closure device (see MIL-P-44146) shall be placed within the outer folds of the mixing bag so as to minimize any sharp protrusions against the pouch.

c. The filled mixing bag shall be folded and fitted into a pouch.

d. The pouch shall be sealed in accordance with the requirements of MIL-P-44146.

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3.4 Finished product. The finished product shall comply with the following requirements:

Dehydrated product:

a. There shall be no foreign material such as but not limited to dirt, glass, paint, hard plastic, wood, insects, insect parts or noxious chemicals. There shall be no innocuous soft extraneous material (paper, string, etc.), smears or stains (oil, grease, etc.).

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. There shall be no evidence of incomplete dehydration.

e. When the product is in the form of slices:

1. Each pouch shall contain four slices.
2. Each slice shall be approximately 2.25 $\pm$ 0.1 by 3.25 $\pm$ 0.1 by 1/2 to 3/4 inches.
3. The volume of the four slices shall be not greater than 246 cubic centimeters.
4. The total weight of the four slices shall be not less than 90 grams and not greater than 100 grams.

f. When the product is in the form of bars:

1. Each pouch shall contain three bars.
2. Each bar shall be 3 $\pm$ 0.1 by 1 $\pm$ 0.1 by not greater than 1-1/4 inches.
3. The total weight of the three bars shall be not less than 90 grams and not greater than 100 grams.
4. Each bar shall have rounded edges and corners.

g. The product shall be sufficiently compressed or formed so that it can be packaged without breaking or crumbling.

h. Moisture content of the dehydrated product shall not exceed 2.0 percent.

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i. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

j. The E. coli count shall be less than 3 per gram (no positives in the standard 3 tube MPN technique) in four of five samples and not greater than 20 per gram in any sample.

Rehydrated product:

k. When the product is rehydrated as specified in footnote 3 of table III the product shall show not less than 75 percent rehydration.

l. The rehydrated escalloped potatoes with pork shall possess flavor, odor and texture considered normal for frozen and heated escalloped potatoes with pork. There shall be no scorched, burnt, or foreign flavors or odors.

m. When rehydrated, the escalloped potatoes with pork shall show distinct pieces of pork meat (approximately 1/2 by 1/2 by 1/4 inch), and potatoes. The product shall not be a mass of indistinguishable materials.

3.4.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The pork component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Pork examination for condition and cut. All pork shall be examined in the bone-in state for conformance to the condition and cut requirements in 3.2.1. Any nonconforming pork shall be rejected.

4.5.1.2 Boned and trimmed pork examination. After boning and trimming and prior to any further processing, the pork shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agricultural Marketing Services (AMS) agent by the contractor.

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However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless pork. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the pork to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for pork rejected because of presence of off-condition, the pork may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Pork shall not be reexamined more than one time.

TABLE I. Sampling and acceptance criteria for boned and trimmed pork

Lot size (pounds)	Sample size (No. sample units)	Defect categories Major AC	Major RE	Minor AC	Minor RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10000	50	5	6	10	11
10001 to 35000	80	7	8	14	15
Reinspection lots of 10001 to 35000	125	10	11	21	22

TABLE II. Boned and trimmed pork defects

Category		Defect 1/ 2/
Major	Minor	
151		Presence of blood clot measuring 0.5 inch or more in any dimension.

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TABLE II. Boned and trimmed pork defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	201	Presence of blood clot measuring 0.3 inch or more but less than 0.5 inch.
152		Presence of bruise measuring 1.0 inch or more in any dimension.
	202	Presence of bruise measuring 0.3 inch or more but less than 1.0 inch in any dimension.
153		Presence of bone measuring 0.3 inch or more in any dimension.
154		Presence of cartilage measuring 0.5 inch or more in any dimension.
	203	Presence of cartilage measuring 0.3 inch or more but less than 0.5 inch in any dimension.
155		Presence of major lymph gland measuring 1.0 inch or more in any dimension.
	204	Presence of exposed lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension.
156		Presence of skin measuring 0.3 inch or more in any dimension.
157		Exposed (protruding) ligaments and tendons measuring more than 0.3 inch.
158		Presence of kidney measuring more than 1.0 inch in any dimension.

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TABLE II. Boned and trimmed pork defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
159		Presence of ham facing.
	205	Presence of shank with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	206	Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension.
	207	Presence of skirt (diaphragm) or abdominal tissue.
	208	Presence of leaf, lumbar or pelvic fat.
	209	Presence of mediastinal tissue in the first rib or sternal region.
	210	Presence of dehydrated surface measuring 1.0 square inch or more.

1/ Acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insects, insect parts, hair, wood, metal) shall be determined by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or presence of any off-condition (including but not restricted to being off-odor, slightly sticky, gassy, rancid, sour, or showing discoloration) shall cause rejection of the involved lot.

4.5.1.3 Other ingredient and component examination. Conformance of ingredients and components to identity, condition and other requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance

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to any identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to preparation, processing, handling and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.2.1 In-process testing for fat content. Four 1-pound samples shall be selected at random from the day's production of raw ground or flake-cut pork. Each of the four samples shall be tested for fat content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists. Results shall be reported to the nearest 1 percent. If the fat content of one or more samples is greater than the limit specified in 3.3.1, the lot of ground or flaked pork or any product made therefrom shall be rejected.

4.5.3 Filled and sealed pouch examination. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44146.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch, mixing bag and closure device. Any individual net weight of less than 90 grams or greater than 100 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 1 gram.

4.5.5 Volume test (slices only). Test for volume of slices shall be performed as follows:

Equipment

a. Container, large enough to submerge pouches in; having a spout approximately 1 inch below the container surface.

b. 150 gm weight with a clip attached.

c. 250 mL or 500 mL beaker.

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d. 250 mL or 500 mL graduated cylinder.

e. Vacuum-sealed pouch containing a mixing bag and a closure device.

Procedure

a. Establish tare as follows:

(1) Place container on a level, vibration-free surface.

(2) Fill the container with water until the water comes out the spout. Wait until the water stops dripping before proceeding.

(3) Position the beaker under the spout.

(4) Attach the vacuum-sealed pouch containing a mixing bag and a closure device to the 150 gm weight with the clip. Submerge the pouch with the attached weight into the filled container.

(5) Allow the water to stop dripping before measuring, approximately 5 minutes.

(6) Measure the collected water in the graduated cylinder and record this measurement as the volume of the tared pouch. Report measurement to the nearest 2 mL.

(7) Empty and dry the beaker and graduated cylinder.

b. Testing of filled and sealed pouch:

(1) Refill the container and, using the same procedure as above, test the filled and sealed pouch.

(2) Measure the collected water in the graduated cylinder and record this measurement as the volume of the filled and sealed pouch. Report measurement to the nearest 2 mL.

c. Determine the volume of the slices using the following equation:

Volume of filled and sealed pouch	minus	Volume of the tared pouch	=	Volume of the slices.

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The lot size shall be expressed in filled and sealed pouches. Any result not conforming to the requirement on 3.4e shall be scored as a major defect. The sample size shall be the number of pouches indicated by inspection level S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.6 Product examination. The finished product shall be examined for the defects listed in table III. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE III. Product defects 1/ 2/

Category		Defect
Major	Minor	
151		Mixing bag missing.
152		Number of bars/slices in pouch not as specified.
	201	Bag closure device missing.
		Applicable to dehydrated product
153		Evidence of incomplete dehydration.
	202	Broken or crumbled (more than 3 large pieces) bar/slice.
	203	Dimensions of compressed bars not as specified.
	204	Edges and corners of compressed bars not rounded.
		Applicable to rehydrated product 3/
154		Product shows less than 75 percent rehydration. 4/

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TABLE III. Product defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
155		Product does not show distinct pieces of pork meat (approximately 1/2 by 1/2 by 1/4 inch), and potatoes (i.e., the product is a mass of indistinguishable materials).

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, noxious chemicals, rancid, sour, stale, etc.), foreign color, or innocuous soft extraneous material, smears or stains (oil, grease, etc.) shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ To perform rehydration examination, the entire contents of the pouch shall be used. Break bars or slices each into five or six pieces of comparable size. Add 180° to 200°F water to the broken product in the amount of 3.3 mL of water for each gram of dehydrated product and weigh. Stir to expose all surfaces to the water. Rehydrate for 5 minutes, stir, and rehydrate for an additional 5 minutes. Immediately after the 10 minute rehydration period, pour contents onto a U.S. Standard No. 40 sieve. Drain for 5 minutes and weigh.

4/ Percent rehydration shall be calculated as follows:

$$\frac{\text{Weight of rehydrated product}}{\text{Weight of dry product plus added water}} \times 100 = \text{Percent rehydration}$$

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4.5.7 Moisture content testing. The finished product shall be tested for moisture content in accordance with the Moisture-Drying in Vacuum Method in the Meat Section of the Meat and Poultry Products Chapter in the Official Methods of Analysis of the Association of Official Analytical Chemists, except that a temperature of 70°C for 16 hours under a pressure of 100 mm of mercury shall be used. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4 h shall be considered a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.8 Microbiological testing. Five filled and sealed pouches of finished product shall be selected at random from the lot regardless of lot size. The contents of each sample pouch shall be tested for aerobic plate count and E. coli in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Microbiological Methods: Section: Examination of Frozen, Chilled, Precooked, or Prepared Foods-Official First Action. The diluent shall be added to each sample of dry product and allowed to stand for 15 minutes before the blending of that sample. Continue as directed by AOAC. Any result not conforming to the requirements of 3.4 (i) and (j) shall be considered a test failure. Any test failure shall be cause for rejection of the lot.

4.5.9 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44146.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44146.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.4.1 and 6.2).

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6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Tempering. Tempering shall not be considered freezing and is defined as product chilled below 20°F and tempered up to dicing temperature.

6.4 Forming of slices. Good results have been achieved for forming the preconcentrated slices by placing the casings into screen molds prior to stuffing. Acceptable molds have been found to be those produced by Beacon Metal Products, Inc. of Chicago, Illinois.

Custodian:

Army - GL

Review activities:

Army - MD, TS
DP - SS

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-44141(GL)	Escalloped Potatoes with Pork, Cooked, Dehydrated (Operational Ration Component)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK Ltr, 1 Apr 84

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